

AI Business Growth Consultant – Project Report

1. Executive Summary

The AI Business Growth Consultant is an end-to-end web application that empowers business owners, marketing teams, and consultants to predict customer churn and monthly revenue, understand why customers churn using SHAP explainability, and receive actionable retention strategies – all through a modern, intuitive dashboard.

Key business outcomes:

- Reduce manual analysis time from days to minutes.
- Identify high-risk customers proactively and intervene with targeted offers.
- Forecast revenue to optimise inventory, marketing, and cash flow.
- Democratise AI – no coding or data science expertise required.

2. Project Architecture

The system is built as a mono-repo with a clear separation of concerns:

ai-business-growth/

```
├── backend/      # Python FastAPI server
│   ├── app/
│   │   ├── api/  # REST endpoints
│   │   ├── core/  # ML pipeline (data_processor, model_trainer, predictor)
│   │   ├── models/ # Pydantic schemas
│   │   └── models/ # Saved .pkl model artifacts
│   └── uploads/   # Uploaded CSV files
│       └── requirements.txt
├── frontend/     # React + Vite + Tailwind
│   ├── src/
│   │   ├── components/ # Reusable UI blocks
│   │   ├── api/        # Axios client
│   │   └── utils/      # PDF report generator
│   └── package.json
```

└─ README.md

Data Flow:

1. User uploads a CSV → stored in `uploads/`.
2. User triggers training → backend cleans, engineers features, trains multiple models (Logistic Regression, Random Forest, XGBoost) and selects the best.
3. User sends customer data for prediction → backend applies pre-trained model and returns probabilities / revenue figures.
4. User requests SHAP explanations → backend computes feature contributions.
5. User requests strategies → backend combines predictions, SHAP, and business rules to output actionable advice.
6. User requests insights → backend aggregates KPIs across the customer base.
7. User downloads a PDF report → frontend collects all data and generates a professional document.

3. Key Features

Feature	Description	Business Benefit
CSV Upload	Drag-and-drop or click to upload customer data (any schema).	Seamless integration with existing spreadsheets.
Auto-ML Training	Trains 3 models per target (churn or revenue) with hyper-parameter tuning.	No manual model selection – always uses the best algorithm.
Churn Prediction	Returns probability (0–1) and binary class (0=stay, 1=churn).	Identify at-risk customers before they leave.
Revenue Forecasting	Returns predicted monthly revenue per customer.	Plan budgets, allocate resources, and upsell high-value segments.
SHAP Explainability	Shows top 3 features driving each prediction (e.g., “ <i>contract_type increases churn by 0.25</i> ”).	Build trust and provide actionable reasons behind predictions.
Strategy Generator	Combines churn probability, revenue tier, and SHAP factors to produce human-readable advice (e.g., “ <i>High risk – send retention offer</i> ”).	Translates numbers into clear next steps for sales/marketing teams.
Dashboard Insights	Aggregates KPIs: total customers, average churn risk, high-risk count, average/total revenue.	Provides a bird’s-eye view of business health.

PDF Report Export	One-click download of a professionally formatted PDF containing Executive Summary, Prediction Table, and Strategy Table with timestamp.	Shareable reports for stakeholders and board meetings.
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4. Technical Implementation

4.1 Backend (Python + FastAPI)

- FastAPI – modern, async web framework with automatic OpenAPI documentation (Swagger).
- Pandas / NumPy – data manipulation and numerical operations.
- Scikit-learn – LogisticRegression, RandomForest, StandardScaler, LabelEncoder.
- XGBoost – gradient boosting for improved accuracy.
- SHAP – TreeExplainer for explainability (fallback to coefficients for linear models).
- Joblib – model serialisation for persistence.
- Uvicorn – ASGI server for production-like development.

4.2 Machine Learning Pipeline

- Data Cleaning – handles missing values (median for numeric, mode for categorical).
- Feature Engineering – creates `tenure\_group`, `charge\_to\_spend\_ratio`, `is\_senior`, etc.
- Encoding – LabelEncoder for categorical columns.
- Scaling – StandardScaler to normalise features.
- Model Selection – trains 3 models per target and selects the one with highest AUC (churn) or R^2 (revenue).
- Error Handling – gracefully falls back to first model if metrics are undefined.

4.3 Frontend (React + Vite + Tailwind)

- React – component-based UI with hooks for state management.
- Vite – fast build tool with hot module replacement.
- Tailwind CSS – utility-first styling for a polished, professional look.
- Axios – HTTP client with request/response interceptors.
- React-dropzone – drag-and-drop file upload.
- Chart.js / react-chartjs-2 – bar charts for churn probabilities.

- jsPDF / jspdf-autotable – PDF generation with tables and custom styling.
- Proxy – Vite proxy forwards `/api` requests to backend during development.

4.4 Error Handling & Resilience

- Backend returns clear, structured error messages (e.g., Churn model not trained).
- Frontend catches all errors and displays user-friendly alerts.
- SHAP explanations gracefully degrade to coefficient-based contributions if Tree Explainer fails.

5. Business Value & Use Cases

Stakeholder	How They Benefit
Startup Founders	Understand which segments drive revenue, forecast cash flow, and reduce churn without hiring a data scientist.
E-commerce Owners	Personalise marketing campaigns, predict inventory needs, and identify high-value customers for loyalty programs.
Marketing Teams	Allocate budget to channels with highest ROI based on churn risk and predicted spend.
Business Consultants	Offer data-driven advice to clients using a white-labelled tool – faster, cheaper, and more scalable than manual analysis.
Students / Job Seekers	Showcase a full-stack AI product with real-world business impact in interviews.

6. Setup & Deployment Instructions

Prerequisites

- Python 3.12+
- Node.js 18+
- Git

Quick Start (Development)

```
```bash
```

Clone the repo

```
git clone <your-repo-url>
```

```
cd ai-business-growth
```

## Backend

```
cd backend
```

```
python -m venv venv
```

```
source venv/bin/activate # or `venv\Scripts\activate` on Windows
```

```
pip install -r requirements.txt
```

```
python -m uvicorn app.main:app --reload
```

## Frontend (in a separate terminal)

```
cd frontend
```

```
npm install
```

```
npm run dev
```

```
'''
```

Visit `http://localhost:3000` for the dashboard.

The API docs are available at `http://localhost:8000/docs`.

## 7. Future Enhancements

- Real-time streaming – connect to live data sources (Shopify, Stripe, HubSpot) via webhooks.
- Multitenancy – support multiple organisations with isolated data and models.
- A/B testing – compare different retention strategies and measure uplift.
- Deep learning – add neural networks for more complex patterns (when dataset size grows).
- Automated retraining – schedule periodic retraining to keep models fresh.
- Customisable thresholds – let users define their own risk levels.

## 8. Conclusion

The AI Business Growth Consultant is a production-ready, full-stack AI application that showcases modern software engineering practices, machine learning ops, and product-centric thinking. It solves a genuine business problem (customer retention) with a user-friendly interface, explainable predictions, and clear actionable outputs – making it an ideal portfolio piece for any data scientist or full-stack developer aiming to demonstrate real-world impact.